

A company wants to find out if a new Machine has improved employee productivity. Before the new Machine, the average productivity was **50 units per day**, and the After the training, a random sample of **30 employees** had an average productivity of **53 units per day**. The company wants to test whether the New machine has significantly increased productivity.

Population mean:	$\mu = 50$
Sample size:	$n = 30$
Sample mean:	$\bar{x} = 53$
Sample standard deviation:	$\sigma = 4.5$
CI	$= 0.97$
Significance Value	$= 0.03$

Define the Null Hypothesis

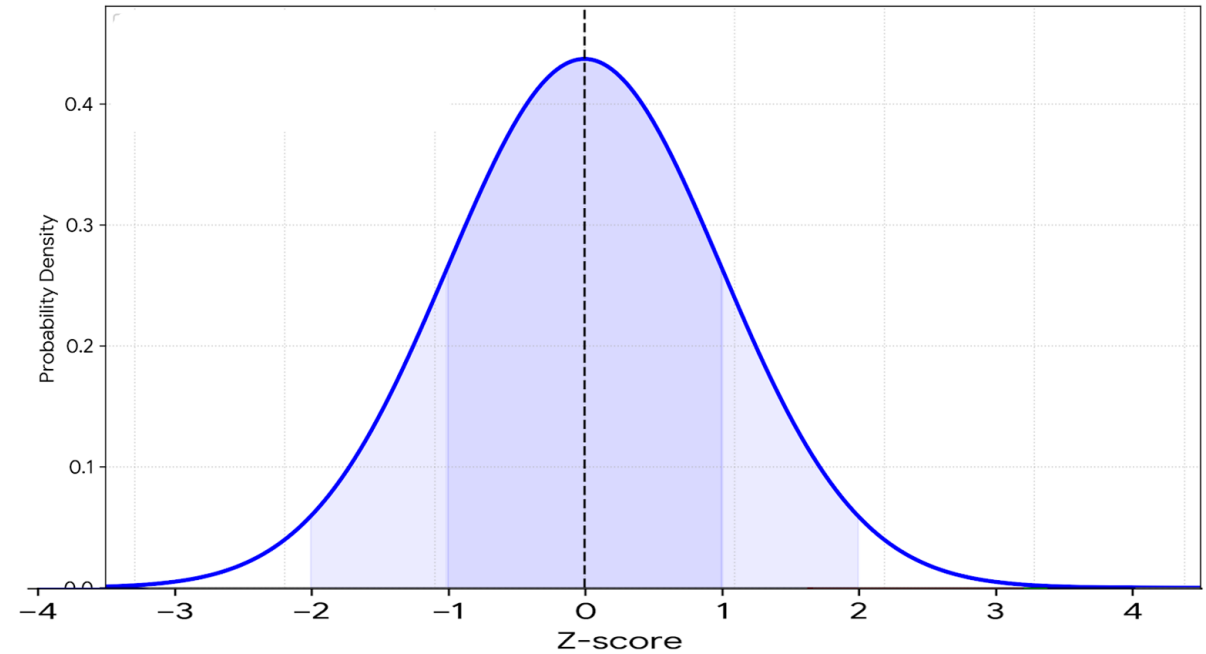
This is a **one-sample right-tailed Z-test** because:

Null Hypothesis (H_0): The New Machine program did not increase productivity.

$H_0 :$ $\mu=50$

Alternative Hypothesis (H_1): New Machine program increased productivity.

$H_1 :$ $\mu>50$



Calculate the

1) **Critical Value for Confidence level(.03) & :**

to know the rejection Area on graph

2) **Z-score :**

to know if the value fall under the reaction area or not

1) **Standard Error :**

Helps to calculate the Z-score.

$$\text{Zscore (Z)} = \frac{\bar{x} - \mu}{SE}$$

$$\text{standard Error (SE)} = \frac{\sigma}{\sqrt{n}}$$

Step: Find Critical Value using t-table

Significance level $\alpha=0.03$

Degree of Freedom(df)

Df = n-1

Df = 30-1

Df = 29

Degree of Freedom(df) = 29

Know the critical values

$t_{0.03, df=29}$

Most t-tables don't directly show 0.03, so we use the closest common values:

0.050 $\rightarrow$ 1.699

0.025 $\rightarrow$ 2.045

Step A: distances in α

From 0.05 $\rightarrow$ 0.03 = 0.02

From 0.05 $\rightarrow$ 0.025 = 0.025

$$\frac{0.05 - 0.03}{0.05 - 0.025} = \frac{0.02}{0.025} = 0.8$$

Difference in t:

$$2.045 - 1.699 = 0.346$$

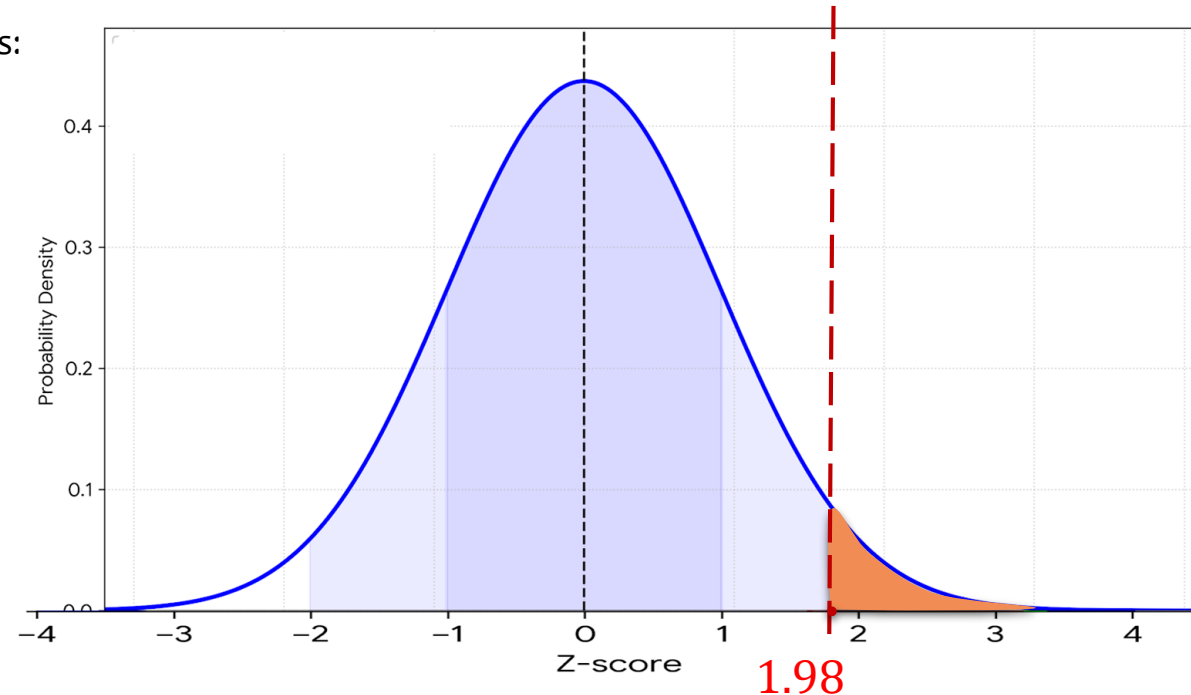
Now take 0.8 of this gap:

$$0.8 \times 0.346 = 0.277$$

Now add to lower value:

$$1.699 + 0.277 = 1.976$$

Critical Value ≈ 1.98



Calculate the standard error

Sample Standard deviation:

$$\sigma = 4.55$$

Sample size:

$$n = 30$$

$$\text{standard Error (SE)} = \frac{\sigma}{\sqrt{n}}$$

$$\text{standard Error (SE)} = \frac{4.5}{\sqrt{30}}$$

$$\text{standard Error (SE)} = \frac{4.5}{5.5772}$$

$$\text{standard Error (SE)} = 0.8215$$

Calculate the Z-Score

Population mean: $\mu = 50$
Standard Error $SE = 0.8215$
Sample mean: $\bar{x} = 53$

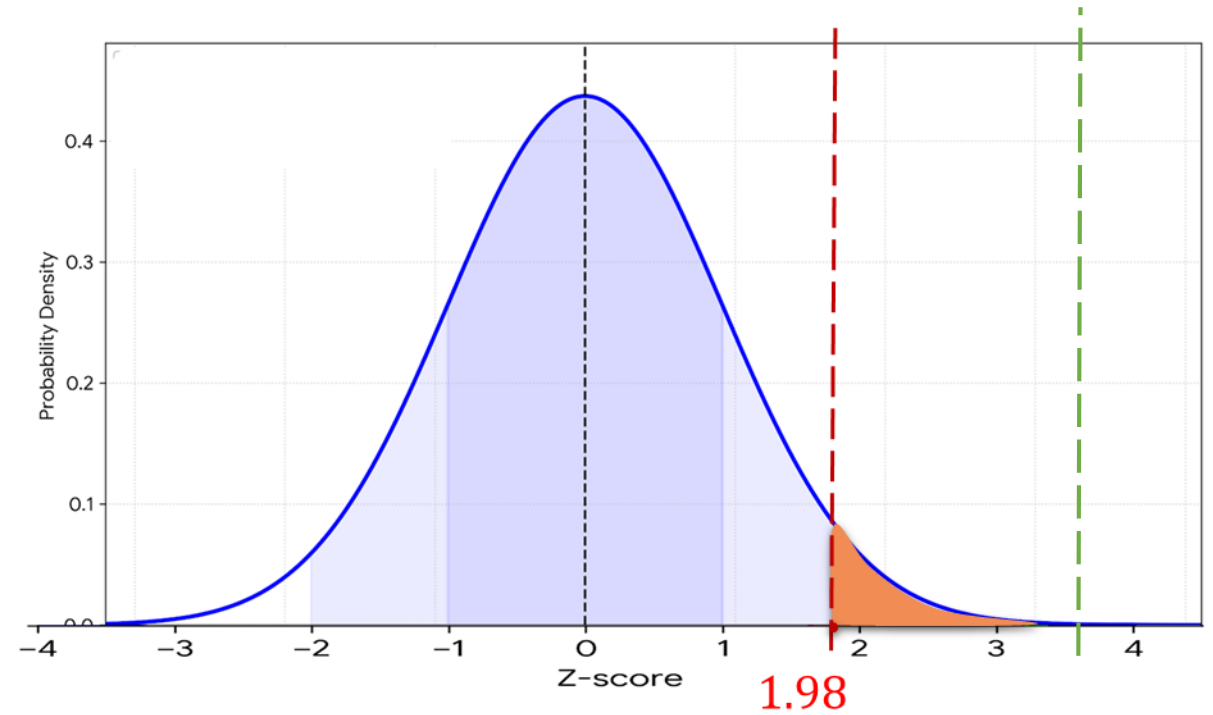
$$\text{Zscore (Z)} = \frac{\bar{x} - \mu}{SE}$$

$$\text{Zscore (Z)} = \frac{\bar{x} - \mu}{SE} = \frac{53 - 50}{0.8216} = \frac{3}{0.8216}$$

$$\text{Zscore (Z)} = 3.65$$

Decision: Reject H_0 because $3.65 > 1.98$.

Conclusion: The New machine increased productivity at the $\alpha = 0.05$ level.



Using p-Value

α	t-value
0.005	2.756
0.001	3.659

$$t=3.65$$

Since 3.65 is closer to the 3.659 we will consider 0.001

$$pvalue(0.001) < \text{significance level}(0.03)$$

We can say that Null hypothesis is rejected

Conclusion

Because the p-value (0.001) is far lower than your significance level (0.03), the result is highly statistically significant.

There is a mere 0.01 chance that this productivity increase happened.

Null hypothesis is rejected.